

COMP702 – Image Processing and Computer Vision

Project – South African Bank Notes Recognition

Due Date: 29/05/2026

- Submit a typed assignment on Moodle
- Submit in **PDF format** not **Zipped files**
- The project can be done in groups of at most **5 students**

This course project is an opportunity for you to explore an Image Processing and Computer Vision problem using a real-world dataset. The idea is to apply multiple Image Processing techniques or major variations of a single approach (typically, different groups will explore different approaches) to compare results and summarise your findings in a report.

The aim of bank note recognition is to identify the properties (features) of the depicted bank note in a reference collection of notes. The bank note recognition system should compute a similarity measure between notes that are invariant to sides, scale, and rotation, because bank notes have two faces and their orientation can be at any angle. Given all the South African bank notes (*both old and new notes*): (**R10, R20, R50, R100, R200**), implement a robust image processing system that can classify each input bank note image appropriately. (Consider both old and new notes, and consider both sides of each note).

This project should be carried out in the following stages (showing all the snapshots):

- Bank Coin Image Preprocessing and Enhancement
- Bank Coin Segmentation
- Bank Coin Features Extraction
- Bank Coin Classification

The above stages should be thoroughly researched, and several algorithms and techniques at each stage should be compared and contrasted, with justification supported by image snapshots from empirical experiments.

The following submissions are required:

- Project report about **4-6 pages** using the following format (*Introduction, Image Preprocessing and Enhancement, Image Segmentation, Feature Extraction, Coin Classification, Conclusion, References*).
- **GitHub link** for the project code (*including a **Help File** on how to run it*).